

AADG classes: FM-index

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1 BWT construction.

Inspect the code for building BWT and answer the following questions:

- What is the worst case time complexity of `bwtViaBwm` function?
- How about the memory complexity? How to improve it?
- What could be the complexity of `bwtFromSA` function if an efficient SA building algorithm was used?

2 BWT reversing.

Inspect the code for reversing BWT and answer the following questions:

- What is the memory complexity of `reverseBwt` function?
- How about time complexity?
- How could it be implemented in a more memory efficient-manner?

3 FM-index.

Inspect the code for the FM-index. Given string `T='alamakotaimapsa'`, use `FmCheckpoints` and `FmIndex` classes to check:

- what is the number of occurrences of `'a'` in `T[:5]` and in `BWT(T)[:5]`,
- whether the string `'chomika'` is a substring of `T`,
- where does the string `'ma'` occur in `T`.

4 Queries with special characters.

Consider the problem of searching patterns with a special symbol **#** that can match any text character (we assume that **#** does not occur in the text).

1. Design the algorithm of FM-index-based searching of patterns with one symbol **#**.
2. Implement your approach as a subclass of the `FmIndex` class from the previous exercise.
3. What is the complexity of your solution?